

Chapter 4: Integers and Rationals

Analysis I, by Terrence Tao

A: The Integers

Problem 1 E

Verify that the definition of equality on the integers is reflexive, symmetric, and transitive.

Proof. First, we show that the equality is reflexive. Consider the integer $a \text{ --- } b$. Notice that $a + b = a + b$, so $a \text{ --- } b = a \text{ --- } b$.

Second, we prove that the equality is symmetric. Suppose $a \text{ --- } b = a' \text{ --- } b'$. Then, $a + b' = a' + b$. Because the equality on the natural numbers is transitive, we have that $a' + b = a + b'$. By definition of the equality on the integers, we have that $a' \text{ --- } b' = a \text{ --- } b$.

Next, we show that the equality on the integers is symmetric. Suppose $a \text{ --- } b = c \text{ --- } d$ and $c \text{ --- } d = e \text{ --- } f$. Then, $a + d = c + b$ and $c + f = e + d$. We add the second equation into the first to get:

$$a + d + c + f = c + b + e + d.$$

By the cancellation law of natural numbers for addition, we can cancel out c and d to get:

$$a + f = b + e.$$

By the definition of equality on the integers, $a \text{ --- } b = e \text{ --- } f$. □

Problem 2 E

Show the definition of negation on the integers is well-defined in the sense that if $(a \text{ --- } b) = (a' \text{ --- } b')$, then $-(a \text{ --- } b) = -(a' \text{ --- } b')$. Hence, equal integers have equal negations.

Proof. Suppose $(a \text{ --- } b) = (a' \text{ --- } b')$, so $a + b' = a' + b$. Because addition of the natural numbers is commutative, we have that $b' + a = b + a'$. Then, by the symmetry of the equality of the natural numbers, we have $b + a' = b' + a$. Then, by the definition of the equality on the integers, we have $b \text{ --- } a = b' \text{ --- } a'$. Because $b \text{ --- } a = -(a \text{ --- } b)$ and $b' \text{ --- } a' = -(a' \text{ --- } b')$, we get:

$$-(a \text{ --- } b) = -(a' \text{ --- } b').$$

□

Problem 3 E

Show that $(-1) \times a = -a$ for every integer a .

Proof. Recall that $-1 = 0 \text{ --- } 1$ and $a = (a \text{ --- } 0)$. Notice:

$$\begin{aligned} (-1) \times a &= (0 \text{ --- } 1) \times (a \text{ --- } 0) \\ &= (0 \times a + 1 \times 0) \text{ --- } (0 \times 0 + 1 \times a) \\ &= (0 + 0) \text{ --- } (0 + a) \\ &= 0 \text{ --- } a \\ &= -a \end{aligned}$$

□

Problem 4 E

Prove the remaining identities in Proposition 4.1.6. For all of the laws of algebra for the integers, let $x = x_1 - x_2, y = y_1 - y_2, z = z_1 - z_2$ be integers, where $x_1, x_2, y_1, y_2, z_1, z_2 \in \mathbb{N}$.

Proposition 4.1. $x + y = y + x$

Proof. Notice:

$$\begin{aligned} x + y &= (x_1 - x_2) + (y_1 - y_2) \\ &= (x_1 + y_1) - (x_2 + y_2) \\ &= (y_1 + x_1) - (y_2 + x_2) \\ &= y + x. \end{aligned}$$

□

Proposition 4.2. $(x + y) + z = x + (y + z)$

Proof. Notice:

$$\begin{aligned} (x + y) + z &= ((x_1 - x_2) + (y_1 - y_2)) \\ &\quad + (z_1 - z_2) \\ &= ((x_1 + y_1) - (x_2 + y_2)) \\ &\quad + (z_1 - z_2) \\ &= (x_1 + y_1 + z_1) - (x_2 + y_2 + z_2) \\ &= (x_1 + (y_1 + z_1)) \\ &\quad - (x_2 + (y_2 + z_2)) \\ &= (x_1 - x_2) \\ &\quad + ((y_1 + z_1) - (y_2 + z_2)) \\ &= x + (y + z). \end{aligned}$$

□

Proposition 4.3. $x + 0 = 0 + x = x$

Proof. Recall that $x = (x - 0)$ and $0 = (0 - 0)$. Notice:

$$\begin{aligned} x + 0 &= (x - 0) + (0 - 0) \\ &= (x + 0) - (0 + 0) \\ &= x - 0 \\ &= x. \end{aligned}$$

Similarly, because addition is commutative, we have that $0 + x = x$. □

Proposition 4.4. $x + (-x) = (-x) + x = 0$

Proof. Notice:

$$\begin{aligned} x + (-x) &= (x_1 - x_2) \\ &\quad + (x_2 - x_1) \\ &= (x_1 + x_2) - (x_2 + x_1) \\ &= (x_1 + x_2) - (x_1 + x_2) \\ &= 0. \end{aligned}$$

Because addition is commutative, we have that $(-x) + x = 0$. □

Proposition 4.5. $xy = yx$

Proof. Notice:

$$\begin{aligned} xy &= (x_1 - x_2) \times (y_1 - y_2) \\ &= (x_1 y_1 + x_2 y_2) - (x_1 y_2 + x_2 y_1) \\ &= (y_1 x_1 + y_2 x_2) - (y_2 x_1 + y_1 x_2) \\ &= (y_1 x_1 + y_2 x_2) - (y_1 x_2 + y_2 x_1) \\ &= yx. \end{aligned}$$

□

Proposition 4.6. $(xy)z = x(yz)$.

Proof. This has already been done in the book. □

Proposition 4.7. $x1 = 1x = x$

Proof. Recall that $1 = (1 - 0)$. Notice:

$$\begin{aligned} 1x &= (1 - 0) \times (x_1 - x_2) \\ &= (1x_1 + 0x_2) - (1x_2 + 0x_1) \\ &= x_1 - x_2 \\ &= x. \end{aligned}$$

Because multiplication is commutative, $x1 = x$. □

Proposition 4.8. $x(y + z) = xy + xz$

Proof. Notice:

$$\begin{aligned} x(y + z) &= (x_1 - x_2)((y_1 - y_2) + (z_1 - z_2)) \\ &= (x_1 - x_2)((y_1 + z_1) - (y_2 + z_2)) \\ &= (x_1(y_1 + z_1) + x_2(y_2 + z_2)) \\ &\quad - (x_1(y_2 + z_2) + x_2(y_1 + z_1)) \\ &= (x_1 y_1 + x_1 z_1 + x_2 y_2 + x_2 z_2) \\ &\quad - (x_1 y_2 + x_1 z_2 + x_2 y_1 + x_2 z_1) \\ &= ((x_1 y_1 + x_2 y_2) + (x_1 z_1 + x_2 z_2)) \\ &\quad - ((x_1 y_2 + x_2 y_1) + (x_1 z_2 + x_2 z_1)) \\ &= ((x_1 y_1 + x_2 y_2) - (x_1 y_2 + x_2 y_1)) \\ &\quad + ((x_1 z_1 + x_2 z_2) - (x_1 z_2 + x_2 z_1)) \\ &= xy + xz \end{aligned}$$

□

Proposition 4.9. $(y + z)x = yx + zx$

Proof. Notice:

$$\begin{aligned} (y + z)x &= x(y + z) \\ &= xy + xz \\ &= yx + zx. \end{aligned}$$

□

Problem 5 H

Let $a, b \in \mathbb{Z}$. Suppose $ab = 0$. Then, $a = 0$ or $b = 0$.

Proof. Suppose that $ab = 0$. We will use the trichotomy of the integers on the variable a and the cancellation law of the naturals. We will prove the proposition for all three cases of $a = 0$, a is a positive integer, and a is a negative integer. In all these cases, b is held fixed with $b = b_1 - b_2$, where $b_1, b_2 \in \mathbb{N}$.

1. Say $a = 0$. Then, the proposition is trivially true.
2. Say a is a positive number, so a takes on the form $a = k - 0$ for some positive natural number k . Then, $(k - 0) \times (b_1 - b_2) = (kb_1 + 0b_2) - (kb_2 + 0b_1) = kb_1 - kb_2$. Because $ab = 0$, we have that $(kb_1 - kb_2) = (0 - 0)$, which means that $kb_1 = kb_2$. By the cancellation law of the naturals, we have that $b_1 = b_2$, so $b = 0$.
3. Say a is a negative number, so a takes on the form $a = 0 - k$. Then, $(0 - k) \times (b_1 - b_2) = (0b_1 + kb_2) - (0b_2 + kb_1) = kb_2 - kb_1$. Because $ab = 0$, we have $(kb_2 - kb_1) = 0$, so $kb_2 = kb_1$. By the cancellation law, we have that $b_2 = b_1$, so $b = 0$.

□

Problem 6 M

Let $a, b, c \in \mathbb{Z}$ with $c \neq 0$. If $ac = bc$, then $a = b$.

There are two ways to prove this. The first proof uses Prop 4.1.8 where it states that integers have no zero divisors.

Proof. Suppose $ac = bc$. Notice:

$$\begin{array}{ll}
 ac - bc = 0 & \text{by subtracting } bc \text{ to both sides} \\
 c(a - b) = 0 & \text{by distributing } c \\
 a - b = 0 & \text{by Prop. 4.1.8 since } c \neq 0 \\
 a = b. & \text{by adding } b \text{ to both sides}
 \end{array}$$

□

The second proof uses the trichotomy of the integers on the variable c , and the cancellation law of multiplication for the natural numbers to eventually remove k in the equations.

Proof. We fix $a = a_1 - a_2$ and $b = b_1 - b_2$. Since $c \neq 0$, we only consider two cases.

1. Say c is a positive number, so there exists a positive natural number k such that $c = k - 0$. Notice:

$$\begin{aligned}
 ac = bc &\implies (a_1 - a_2)(k - 0) = (b_1 - b_2)(k - 0) \\
 &\implies ka_1 - ka_2 = kb_1 - kb_2 \\
 &\implies ka_1 + kb_2 = kb_1 + ka_2 \\
 &\implies k(a_1 + b_2) = k(b_1 + a_2) \\
 &\implies a_1 + b_2 = b_1 + a_2 \\
 &\implies a_1 - a_2 = b_1 - b_2 \\
 &\implies a = b.
 \end{aligned}$$

2. Say c is a negative number, so c is the negation $-k$ of a positive natural number k . In other words, $c = 0 - k$. We get a very similar proof to the second case of c being a positive number. Notice:

$$\begin{aligned}
 ac = bc &\implies (a_1 - a_2)(0 - k) = (b_1 - b_2)(0 - k) \\
 &\implies ka_2 - ka_1 = kb_2 - kb_1 \\
 &\implies k(a_2 + b_1) = k(b_2 + a_1) \\
 &\implies a_2 + b_1 = b_2 + a_1 \\
 &\implies a_1 - a_2 = b_1 - b_2 \\
 &\implies a = b.
 \end{aligned}$$

□

Lemma 1. Let $a, b, k \in \mathbb{Z}$. If $a = b + k$, THEN $a = b$ if and only if $k = 0$.

Proof. Suppose $a = b + k$. Let us prove the forward direction. Assume $a = b$. Then $a = b = b + k$. We subtract b to all sides to get $0 = k$. To prove the converse, assume that $k = 0$. Trivially, $a = b + k = b + 0 = b$. □

Problem 7 E

Prove the properties of order for the integers. For all the propositions, let $a, b, c \in \mathbb{Z}$.

Proposition 7.1. $a > b$ if and only if $a - b$ is a positive number.

Proof. Suppose $a > b$. By definition of order, there exists some natural number k such that $a = b + k$ and $a \neq b$. Hence, by Lemma 1, $k \neq 0$, and k therefore is a positive number. In the other direction, suppose $a - b$ is a positive number. Define $k := a - b \neq 0$, so $a = b + k$, and hence $a > b$. By Lemma 1, since $k \neq 0$, we have that $a \neq b$. Therefore, $a > b$. □

Proposition 7.2. Addition preserves order. In other words, if $a > b$, then $a + c > b + c$.

Proof. Suppose $a > b$. Then, $a = b + k$ for some positive number k . Adding c to both sides of the equation gives $(a + c) = (b + c) + k$. Hence, $a + c > b + c$. □

Proposition 7.3. Positive multiplication preserves order. If $a > b$ and c is positive, then $ac > bc$.

Proof. Suppose $a > b$, so $a = b + k$ for some positive number k . Notice $ac = (b + k)c = bc + kc$. Since $c \neq 0$ and $k \neq 0$, we have $kc \neq 0$. Therefore, $ac > bc$. □

Proposition 7.4. Negation reverses order. If $a > b$, then $-a < -b$ (i.e., $-b > -a$).

Proof. Suppose $a > b$, so $a = b + k$ for some positive number k . We take the negation to get $-a = -b - k$. Notice that $-b = -a + k$, so $-b > -a$. □

Proposition 7.5. Order is transitive. If $a > b$ and $b > c$, then $a > c$.

Proof. Suppose $a > b$ and $b > c$, so $a = b + k$ and $b = c + l$ for positive numbers k and l . Notice, $a = b + k = c + l + k = c + (l + k)$. Since $l + k$ is also a positive number, we have $a > c$. □

Proposition 7.6. Order is trichotomous. Exactly one of the statements hold $a > b$, $a < b$, or $a = b$ is true.

Proof. Consider any two integers a and b . By the trichotomy of the integers, we have exactly one of these to be true: $a - b > 0$, $a - b < 0$, or $a - b = 0$. We consider each case and how it relates to the proposition.

1. Say $a - b > 0$, so $0 + k = (a - b)$ for some positive number k . By rearrangement, $b + k = a$, so $a > b$.
2. Say $a - b < 0$, so $(a - b) + l = 0$ for some positive number l . By rearrangement, $a + l = b$, so $a < b$.
3. Say $a - b = 0$. Then, $a = b$.

□

Problem 8 M

Show that the principle of induction does not apply directly to the integers. More precisely, given an example of a property $P(n)$ pertaining to an integer n such that $P(0)$ is true, and that $P(n)$ implies $P(n++)$ for all integers n , but that $P(n)$ is not true for all integers n . Thus, induction is not as useful a tool for dealing with the integers as it is with the natural numbers.

Proof. Consider the property $P(n) : n \geq 0$. Trivially, $P(0)$ is true. To show that the inductive hypothesis holds, suppose that $P(n)$ is true for any integer n , so $n \geq 0$. Clearly, $n++ \geq 0$, so $P(n++)$ holds true. However, $P(-1)$ is false.

The more interesting question though posed by Issa Rice is “What must the solution space look like?” More precisely, we can ask “Is there some simple characterization of every property P that works as a solution to this exercise?” The answer is that for any property P , its solution space will look the same as the set of all integers greater than some non-positive integer m . In other words, $\{n \in \mathbb{Z} : P(n)\} = \{n \in \mathbb{Z} : n \geq m\}$ for some $m \leq 0$. □

B: The Rationals

Problem 1 E

Verify that the definition of equality on the rationals is reflexive, symmetric, and transitive.

Proof.

□

Problem 2 E

Let $x, y, z \in \mathbb{Q}$. Then, the following properties hold:

Proposition 2.1. $x + y = y + x$

Proof. Chuchu

□

Problem 3 H

Let $x \in \mathbb{Q}$. We show the trichotomy of $\mathbb{Q}$, where exactly one of the following statements hold: $x = 0$, x is a positive rational number, or x is a negative rational number.

Proof. shcu

□

Problem 4 A

For the properties of the order on the rationals, let $x, y, z \in \mathbb{Q}$. Then, the following properties hold.

Problem 5 H

Show that if x, y, z are rational numbers such that $x < y$ and z is negative, then $xz > yz$.

Proof. chucu

□

C: Absolute Value and Exponentials

Problem 1 E

Verify the basic properties of absolute value and distance. Let $x, y, z \in \mathbb{Q}$.

Proposition 1.1. *Non-degeneracy of absolute value.* We have $|x| \geq 0$. Also, $|x| = 0$ if and only if $x = 0$.

Proof. d

□

Proposition 1.2. *(Triangle inequality for absolute value)* We have $|x + y| \leq |x| + |y|$.

Proof. s

□

Proposition 1.3. We have the inequalities $-y \leq x \leq y$ if and only if $y \geq |x|$. In particular, we have $-|x| \leq x \leq |x|$.

Proof. s

□

Proposition 1.4. *(Multiplicativity of absolute value)* We have $|xy| = |x||y|$. In particular, $|-x| = |x|$.

Proof. s

□

Proposition 1.5. *(Non-degeneracy of distance)* We have $d(x, y) \geq 0$. Also, $d(x, y) = 0$ if and only if $x = y$.

Proof. smth

□

Proposition 1.6. *(Symmetry of distance)* We have $d(x, y) = d(y, x)$.

Proof. sh

□

Proposition 1.7. *(Triangle inequality for distance)* We have $d(x, z) \leq d(x, y) + d(y, z)$.

Proof. chu

□

Problem 2 E

Let $x, y, z, w \in \mathbb{Q}$. Then, the following properties hold:

Proposition 2.1. The number x is ε -close to y for every $\varepsilon > 0$ if and only if $x = y$. *Proof.* chuchu □

Proof. s □

Proposition 2.2. Let $\varepsilon > 0$. If x is ε -close to y , then y is ε -close to x .

Proof. s □

Proposition 2.3. Let $\varepsilon, \delta > 0$. If x is ε -close to y , and y is δ -close to z , then x and z are $(\varepsilon + \delta)$ -close.

Proof. chuchu □

Proposition 2.4. Let $\varepsilon > 0$. If x and y are ε -close, they are also ε' -close for every $\varepsilon' > \varepsilon$.

Proposition 2.5. Let $\varepsilon > 0$. If y and z are both ε -close to x and w is between y and z (i.e., $y \leq w \leq z$ or $z \leq w \leq y$), then w is also ε -close to x .

Proof. chuchu □

Proposition 2.6. Let $\varepsilon > 0$. If x and y are ε -close, and z is non-zero, then xz and yz are $\varepsilon|z|$ -close.

Proof. □

Proposition 2.7. Let $\varepsilon, \delta > 0$. Suppose x and y are ε -close, and z and w are δ -close. Then xz and yw are $(\varepsilon|z| + \delta|x| + \varepsilon\delta)$ -close.

Problem 3 E

(Properties of Exponentiation I). Let x, y be rational numbers. Let n, m be natural numbers. The following properties hold.

Proposition 3.1. We have $x^n x^m = x^{n+m}$. *Proof.* suppose □

Proof. chuchu □

Proposition 3.2. We have $(x^n)^m = x^{nm}$. *Proof.* suppose □

Proof. chuchu □

Proposition 3.3. $(xy)^n = x^n y^n$. *Proof.* suppose □

Proof. chuchu □

Proposition 3.4. Suppose $n \geq 0$. Then, we have $x^n = 0$ if and only if $x = 0$. *Proof.* suppose □

Proposition 3.7. We have $|x^n| = |x|^n$.

Proof. suppose □

Problem 4 E

(Properties of Exponentiation I). Let x, y be rational numbers. Let n, m be integers. The following properties hold.

Proposition 4.1. We have $x^n x^m = x^{n+m}$. *Proof.* suppose □

Proof. chuchu □

Proposition 4.2. We have $(x^n)^m = x^{nm}$. *Proof.* chuchu □

Proof. chuchu □

Proposition 4.3. $(xy)^n = x^n y^n$. *Proof.* suppose □

Proof. chuchu □

Proposition 4.4. If $x \geq y > 0$, then $x^n \geq y^n > 0$ if n is positive OR $0 < x^n \leq y^n$ if n is negative. *Proof.* suppose □

Proposition 4.6. We have $|x^n| = |x|^n$.

Proof. suppose □

Problem 5 E

Prove that $2^N \geq N$ for all positive integers N .

Proof. chuchu □